

ST-MTLNet: Spatio-Temporal Representations of Points of Interest for Multi-Task Learning

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Abstract. *This work proposes ST-MTLNet, a multi-task architecture for POI Category Classification and Next-POI Prediction based on decoupled representations. The model combines a continuous spatial encoder for geographic coordinates, a temporal encoder (Time2Vec) for visitation patterns, and a hierarchical categorical encoder (HGI) for structural and regional POI context. Two spatial encoders, SIREN and Sphere2Vec-M, originally proposed for remote sensing and ecology tasks, are evaluated in the context of multi-task POI modeling on LBSNs. Experiments on the Gowalla dataset across Florida, California, and Texas show that the proposed approach outperforms the baseline in all 21 category-state combinations for classification, with average gains of 20–24 percentage points, and in 76% of combinations for next-POI prediction. The comparison between spatial encoders also reveals complementary performance profiles associated with the geographic distribution of POIs in each territory.*

Resumo. *Este trabalho propõe o ST-MTLNet, uma arquitetura multitarefa para classificação de categoria de POI e previsão do próximo POI baseada em representações desacopladas. O modelo combina uma representação espacial contínua para coordenadas geográficas, uma representação temporal (Time2Vec) para padrões de visitação e uma representação categórica hierárquica (HGI) para contexto estrutural e regional dos POIs. Duas arquiteturas de codificação espacial, SIREN e Sphere2Vec-M, originalmente propostas para sensoriamento remoto e ecologia, são avaliadas no contexto de tarefas multitarefa de POIs em LBSNs. Experimentos com o dataset Gowalla nos estados da Flórida, Califórnia e Texas demonstram que a abordagem proposta supera o baseline em todas as 21 combinações de categoria e estado para classificação, com ganhos médios de 20 a 24 pontos percentuais, e em 76% das combinações para previsão do próximo POI. A comparação entre as arquiteturas espaciais revela ainda perfis complementares de desempenho associados à distribuição geográfica dos POIs em cada território.*

1. Introduction

Location-Based Social Networks (LBSNs), such as Gowalla [Cho et al. 2011, Jure 2014], record user *check-ins* with geographic location, visit time, and category of the visited place. In this context, a Point of Interest (POI) corresponds to a semantically identifiable physical place, such as restaurants, stores, airports, or parks. The analysis of these data is relevant to applications such as recommendation and urban mobility. In this scenario, two tasks stand out:

1. **POI Category Classification:** classify the semantic category (e.g., *Food, Shopping, Travel*) of a POI from its features.

2. **Next-POI Prediction:** predict the category of the next POI that a user will visit, given the historical sequence of their *check-ins*.

In principle, these tasks may be treated as related, since they share the same mobility dataset and the same semantic space of categories, which makes multi-task learning (MTL) a promising strategy to exploit information shared between them [Caruana 1997]. However, they impose distinct demands. POI Category Classification is a non-sequential task, since it depends mainly on the attributes of the POI itself and on its spatial context. Next-POI Prediction is explicitly sequential, requiring the modeling of temporal order, recurrence, and transitions along the user trajectory. Thus, the information useful for one task may not benefit the other in the same way.

MTLNet, proposed in [Silva et al. 2025], exploits this potential by modeling the two tasks in a single multi-task architecture with hard parameter sharing. However, the model uses exclusively vector representations based on the graph structure, generated by *Deep Graph Infomax* (DGI), which encodes spatial and categorical information in a single 64-dimensional vector, without explicitly incorporating continuous geographic coordinates or temporal visitation patterns.

Given this, the question arises of whether decomposing the input into specialized *encoders* for space, time, and category produces a base more suitable for the two tasks than the monolithic *embedding* of DGI. The hypothesis is that decoupled representations capture complementary dimensions of human mobility that are not explicitly modeled by the original approach.

In this sense, this work proposes ST-MTLNet (*Spatial-Temporal MTLNet*), which integrates three independent representations: a continuous spatial representation for geographic coordinates, a temporal representation (Time2Vec) for visitation patterns, and a hierarchical categorical representation for regional and structural relationships between POIs. In the spatial dimension, we evaluate two architectures with distinct assumptions, SIREN and Sphere2Vec-M, originally validated in geospatial tasks of remote sensing and ecology [Wu et al. 2024], but still little explored in multi-task learning of POIs in LBSNs.

The experiments were carried out with the public Gowalla dataset [Jure 2014] in the states of Florida, California, and Texas. In categorical classification, ST-MTLNet outperforms MTLNet in all evaluated category-state combinations, with average gains of 20 to 24 percentage points. In Next-POI Prediction, the proposed model wins in most scenarios, with emphasis on categories such as *Food*. The comparison between SIREN and Sphere2Vec-M also shows that, although the modular approach is consistently superior to the *baseline*, the most suitable spatial *encoder* depends on the geographic distribution of the POIs in each territory.

Therefore, the main contributions of this work are:

- The proposal of ST-MTLNet, which incorporates continuous spatial, temporal, and hierarchical categorical representations into MTLNet.
- The comparative evaluation of SIREN and Sphere2Vec-M in three states with distinct territorial structures.
- The public release of the source code and the experimental *pipeline*.¹

This work is organized as follows: Section 2 presents the related work. Section 3 describes the proposed methodology. Section 4 discusses the experimental results. Finally, Section 5 presents the conclusions and future work.

¹The complete source code is available at https://github.com/TarikSalles/Spatial_Embeddings

2. Related Work

POI prediction and classification are fundamental challenges in location-based recommendation systems and in urban mobility analysis. This section organizes the related work into four axes: graph-based representations, spatial encoders, temporal representations, and multi-task learning applied to POIs.

2.1. Graph-Based POI Representations

Modeling POIs through graphs allows capturing spatial and semantic neighborhood relationships between places. The POI2Vec model [Feng et al. 2017] adapts the Word2Vec architecture to generate POI *embeddings* that incorporate the geographic influence between places through a geographic binary tree structure. CATAPE (*Category-Aware Location Embedding*) [Rahmani et al. 2019] extends this idea by incorporating categorical and sequential information, capturing the geographic influence between POIs based on the temporal sequence of user visits.

In a self-supervised approach, *Deep Graph Infomax* (DGI) [Veličković et al. 2019] maximizes the mutual information between local (node) and global (graph) representations to generate *embeddings* without explicit supervision. HGI (*Hierarchical Graph Infomax*) [Huang et al. 2023] advances this line by operating at multiple hierarchical levels (POI, region, and city), producing representations enriched with regional information through graph convolutions and multi-attention mechanisms.

However, these approaches encode spatial information implicitly through the graph topology, without directly using the geographic coordinates as an input signal for the representation.

2.2. Spatial Representations

The direct use of geographic coordinates to generate latent spatial features is an approach distinct from graph-based representations. The TorchSpatial *framework* [Wu et al. 2024] establishes a unified *benchmark* for evaluating spatial representation strategies, allowing systematic comparison between different architectures.

The SIREN methodology (*Sinusoidal Representation Networks*) [Sitzmann et al. 2020], applied to the geographic context [Rußwurm et al. 2024], models continuous functions through sinusoidal activations, being able to represent high-frequency signals in spatial data. Sphere2Vec [Mai et al. 2023] operates directly on spherical coordinates with multiple resolution scales, preserving geodesic distance properties that are lost in planar projections.

Although these strategies have been evaluated on geospatial tasks such as species prediction and population estimation, their application can be extended to POI prediction and classification models as proposed in this work.

2.3. Temporal Representations

The temporal dimension is central to POI prediction tasks, since user visitation patterns present regularities (e.g., meal times, weekly movements). The LSTPM model (*Long- and Short-Term Preference Modeling*) [Sun et al. 2020] captures short- and long-term preferences using a Geo-Dilated RNN that relates non-consecutive POIs through temporal sequences of visits and geographic information. TransTARec [Sun 2024] proposes an adaptive temporal

translation model that unites representations of the POI, the *timestamp*, and user preferences for Next-POI Prediction.

These models incorporate the temporal dimension mainly in the context of the user’s visit sequence. In this sense, Time2Vec [Kazemi et al. 2019] proposes a temporal representation that combines linear terms with sinusoidal functions of learnable frequency and phase, capturing periodic and non-periodic patterns from continuous temporal values.

2.4. Multi-Task Learning for POI Tasks

Multi-task learning (MTL) [Caruana 1997] has been explored in POI tasks mainly to combine Next-POI Prediction with temporal, behavioral, or regional signals. MCARNN [Liao et al. 2018] integrates sequential dependencies and temporal regularities to simultaneously predict future activities and places. MTPR [Xia et al. 2020] jointly models location and temporal context through geographic LSTMs and adversarial learning. In a more recent line, Halder et al. [Halder et al. 2022] propose a multi-task model based on *Transformers* to recommend the next POI and simultaneously predict the waiting time.

In categorical classification, TME [Xu et al. 2023] explores a hierarchical category structure for semantic annotation of POIs, while HMT-GRN [Lim et al. 2022] combines Next-POI Prediction and prediction of its geographic region in a recurrent model with hierarchical graphs. Together, these works indicate that MTL is promising in POI problems, but there is still little exploration of architectures that jointly address *POI Category Classification* and *Next-POI Prediction* with continuous spatial and explicit temporal representations. This gap motivates the investigation proposed in this work.

3. Methodology

As discussed in Section 2, MTLNet [Silva et al. 2025] uses a single vector $\mathbf{E}_{DGI} \in \mathbb{R}^{64}$ as input for both tasks, jointly encoding spatial and categorical information through the graph topology. One limitation of this monolithic approach is that it does not incorporate continuous geographic coordinates or temporal visitation patterns, factors that, as shown by works on spatial [Wu et al. 2024] and temporal [Kazemi et al. 2019] encoding, carry information complementary to the topological relationships of the graph.

The central design decision of this work is to decompose the unimodal DGI representation into three independent 64-dimensional components, trained separately and integrated by concatenation: a continuous spatial *encoder*, a temporal *encoder* (Time2Vec), and a hierarchical categorical *encoder* (HGI). The underlying hypothesis is that a decoupled representation, with specialized *encoders* trained with loss functions suited to each dimension of the information, produces more expressive representations than a unimodal representation that encodes all dimensions jointly. To evaluate this hypothesis and investigate the impact of the spatial encoding strategy, two *encoders* with distinct architectural assumptions are compared: SIREN and Sphere2Vec-M.

The proposed methodology is illustrated in Figure 1. The following describes the MTLNet model, the *baseline* with DGI, the rationale and operation of each proposed *encoder*, and the strategy for integrating the *embeddings*.

MTLNet [Silva et al. 2025] is a multi-task network built on hard parameter sharing [Caruana 1997], which simultaneously processes the POI Category Classification and Next-POI Prediction tasks. The architecture uses task-specific *encoders* implemented as MLPs with LayerNorm and Dropout, which project the input *embeddings* into a shared latent space

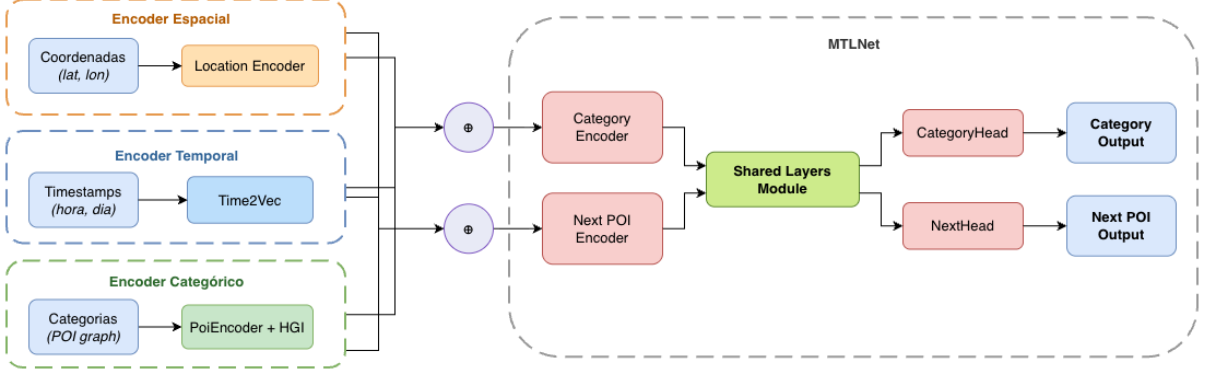


Figure 1. Architecture based on MTLNet [Silva et al. 2025]. The spatial, temporal, and categorical *encoders* are trained in a decoupled manner and generate their respective *embeddings*, which are integrated as input to the model and processed by the model’s shared layers (*Shared Layers Module*) and by the task-specific layers, producing the *Category Output* and *Next POI Output*.

of dimension $d_{\text{shared}} = 256$. Next, FiLM modulation (*Feature-wise Linear Modulation*) [Perez et al. 2018] is applied, conditioned by a task-identifier *embedding* $\mathbf{e}_t$, which generates scale γ_t and shift β_t parameters to modulate the encoded representations:

$$\text{FiLM}(\mathbf{h}_{\text{enc}}^{(t)} | \mathbf{e}_t) = \gamma_t \odot \mathbf{h}_{\text{enc}}^{(t)} + \beta_t \quad (1)$$

The modulated representations are processed by four shared residual blocks with Layer-Norm, LeakyReLU, and Dropout, allowing the model to learn a common representation for both tasks [Baxter 2000].

At the output, each task has a specialized *head*. The category classification module uses a set of three parallel MLPs with varying depths (2, 3, and 4 layers), whose outputs are concatenated and projected to the 7 POI classes. The next-POI *head* employs a *Transformer encoder* with 8 attention *heads* and 4 layers, a causal mask for autoregressive processing, and attention-weighted pooling using attention weights learned over the *timesteps* of the input sequence.

Multi-task training uses the Nash-MTL regularizer [Navon et al. 2022], which formulates gradient balancing as a cooperative Nash bargaining game among K tasks. Given the gradients of each task, Nash-MTL seeks the update direction that maximizes the product of the utilities of all tasks, which ensures that the update is beneficial for all tasks simultaneously, avoiding the dominance of one task over the other.

3.1. Baseline: MTLNet with DGI

The *baseline* model corresponds to the original MTLNet proposed in [Silva et al. 2025], whose internal architecture is kept unchanged in this work. In this model, each POI is represented by a single monolithic *embedding* $\mathbf{E}_{\text{DGI}} \in \mathbb{R}^{64}$, obtained via *Deep Graph Infomax* (DGI) [Veličković et al. 2019] over a spatial graph built by Delaunay triangulation and categorical node attributes, as described in detail in [Silva et al. 2025]. Thus, DGI jointly encodes spatial and categorical relationships in a single vector, without incorporating explicit temporal components. In contrast, the proposed approach replaces this monolithic representation with the concatenation of three specialized *encoders*, totaling 192 input dimensions.

3.2. Data Preparation

For the *POI Category Classification* task, each POI is represented by the *embedding* resulting from the concatenation of the three components, $\mathbf{E}_{cat} = [\mathbf{E}_{HGI} \parallel \mathbf{E}_{loc} \parallel \mathbf{E}_{time}] \in \mathbb{R}^{192}$, forming pairs $(\mathbf{E}_{cat}, c)$ where c is the real category of the POI. In the case of the *baseline*, $\mathbf{E}_{DGI} \in \mathbb{R}^{64}$ is used directly.

For the *Next-POI Prediction* task, the *check-ins* are ordered chronologically per user, and users with fewer than five visits are discarded. Non-overlapping windows of size $L_h = 9$ are extracted: the *check-ins* $(p_1, \dots, p_9)$ form the context, and p_{target} is the next POI whose category must be predicted. Each *check-in* in the window is represented by the corresponding *embedding*, resulting in an input of $9 \times 192 = 1728$ dimensions for ST-MTLNet and $9 \times 64 = 576$ dimensions for the *baseline*. Shorter sequences are filled with padding of zero vectors.

3.3. Spatial Encoder

The original MTLNet model encodes spatial information implicitly through the topology of the Delaunay graph, without directly using geographic coordinates as an input signal. However, as discussed in Section 2, continuous spatial encoders have shown effectiveness in various geospatial tasks [Wu et al. 2024], although their application in multi-task POI models remains little explored. This work selects two *encoders* that represent distinct spatial encoding paradigms: SIREN [Rußwurm et al. 2024], which models continuous functions through sinusoidal activations with controllable frequencies, and Sphere2Vec-M [Mai et al. 2023], which operates directly on spherical coordinates preserving geodesic distance properties. This diversity allows evaluating how different geometric and architectural assumptions impact performance on POI tasks.

All *encoders* produce $\mathbf{E}_{loc} \in \mathbb{R}^{64}$ and are trained with the same binary contrastive loss based on the geographic distance between pairs of coordinates:

$$\mathcal{L}_{loc}(i, j) = - \left[y_{ij} \log \sigma \left(\frac{\text{sim}(z_i, z_j)}{\tau} \right) + (1 - y_{ij}) \log \left(1 - \sigma \left(\frac{\text{sim}(z_i, z_j)}{\tau} \right) \right) \right] \quad (2)$$

in which σ denotes the sigmoid function, $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and $y_{ij} \in \{0, 1\}$ indicates whether the pair is positive ($\Delta d_{ij} \leq 10$ km) or negative ($\Delta d_{ij} \geq 70$ km), with temperature $\tau = 0.15$. Using the same loss function for the two spatial *encoders* allows a controlled comparison, isolating the effect of the encoding architecture.

3.3.1. SIREN

The SIREN model (*Sinusoidal Representation Networks*) [Rußwurm et al. 2024] models a continuous function $f_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^{64}$ that directly maps normalized geographic coordinates into a vector *embedding*. Each layer of the network uses sinusoidal activations with controllable frequencies, allowing high-frequency signals in spatial data to be represented. The output is the *embedding* $\mathbf{E}_{loc} \in \mathbb{R}^{64}$.

3.3.2. Sphere2Vec-M

Sphere2Vec-M [Mai et al. 2023] is a multi-scale encoder formulated to operate directly on spherical coordinates (λ, ϕ) , generating representations that preserve distance properties on

the surface of the sphere. The *sphereM* variant is used, in which $S = 16$ multi-resolution scales are employed with geometrically spaced factors between 10 km and 10,000 km. Interaction terms between λ and ϕ are constructed by keeping one of the components at the highest scale level, allowing the angular resolution of latitude and longitude to be controlled independently. The resulting vector is projected by an MLP to produce $\mathbf{E}_{loc} \in \mathbb{R}^{64}$.

3.4. Temporal Encoder

Human mobility patterns exhibit cyclical regularities, such as meal times and weekly movements, which carry discriminative information about the functional nature of the visited POIs [Sun et al. 2020]. However, the original MTLNet model does not incorporate explicit temporal representations to capture this information.

To fill this gap, Time2Vec [Kazemi et al. 2019] is used, a generic and decoupled temporal representation capable of capturing periodic and non-periodic patterns without the need for manual feature engineering. The architecture combines linear and periodic terms to map the temporal values of each *check-in* (hour of day and day of week, in a coupled manner) into an *embedding* vector. For an *embedding* of dimension D , the function is defined as:

$$\mathbf{f}(\tau)[i] = \begin{cases} \omega_0\tau + \phi_0 & \text{if } i = 0 \\ \sin(\omega_i\tau + \phi_i) & \text{if } 1 \leq i \leq D - 1 \end{cases} \quad (3)$$

in which ω_i and ϕ_i are trainable parameters (weights and *bias*) that capture the periodicity of the temporal visitation patterns. The linear term ($i = 0$) captures global temporal trends, while the sinusoidal terms model cyclical patterns at different frequencies.

The model is trained using the discrete temporal values (hour of day and day of week) in a coupled manner for each *check-in*, with the same binary contrastive formulation employed in the spatial encoder, denoted $\mathcal{L}_{tempo}$. The final output is the *embedding* $\mathbf{E}_{time} \in \mathbb{R}^{64}$, which represents the *timestamp* of each *check-in*.

3.5. Categorical Encoder

The DGI used in the original model encodes categorical information through a *one-hot* matrix of 7 classes processed by the graph structure, without explicitly capturing hierarchical or regional relationships among POI categories. To enrich this dimension, the generation of categorical *embeddings* is carried out in two complementary phases. The POI Encoder learns vector representations that capture co-occurrences among categories from random walks on the spatial graph, and the HGI model [Huang et al. 2023] incorporates hierarchical information at multiple levels (POI, region, and city), producing representations that reflect both the local and the regional context.

3.5.1. POI Encoder

The POI Encoder is trained independently to generate the category *embedding* $\mathbf{E}_{POI_category} \in \mathbb{R}^{64}$. The POIs are organized into a spatial graph built by Delaunay triangulation over the geographic coordinates, with edges weighted by logarithmic decay of the Haversine distance and penalization for connections between distinct counties (GEOIDs).

Over this graph, random walks (*random walks*) are executed following the Node2Vec methodology [Grover and Leskovec 2016]. Each walk is converted into a sequence of se-

condary categories (*fclass*), resulting in local and global co-occurrences among categories. The model learns the *embeddings* using the *skip-gram* strategy with *negative sampling* [Church 2017]:

$$\mathcal{L}_{\text{SGNS}}(i, j) = -\log \sigma(\mathbf{e}_j^\top \mathbf{e}_i) - \sum_{k=1}^K \log \sigma(-\mathbf{e}_{n_k}^\top \mathbf{e}_i),$$

in which $\mathbf{e}_i$ is the *embedding* of the target class, $\mathbf{e}_j$ the positive context *embedding*, and $\mathbf{e}_{n_k}$ represent the negative samples. The implementation incorporates a hierarchical regularization term [Belkin and Niyogi 2003] between category and *fclass*: $\mathcal{L}_{\text{hier}} = \sum_{(c,s) \in \mathcal{H}} \|\mathbf{e}_s - \mathbf{e}_c\|_2^2$, in which $\mathcal{H}$ contains the (category, *fclass*) pairs. The total loss is $\mathcal{L} = \mathcal{L}_{\text{SGNS}} + \lambda_{\text{hier}} \mathcal{L}_{\text{hier}}$. In the end, the resulting *embedding* is generated per category and remapped to each POI.

3.5.2. Hierarchical Graph Infomax (HGI)

After the generation of the *embeddings* $\mathbf{E}_{\text{POI-category}}$, they are used as input to the Hierarchical Graph Infomax (HGI) model [Huang et al. 2023], which enriches the representation by incorporating regional geographic information. The model optimizes mutual information at two hierarchical levels (POI-region and region-city):

$$\mathcal{L} = \alpha \mathcal{L}_{\text{POI-region}} + (1 - \alpha) \mathcal{L}_{\text{region-city}},$$

in which $\alpha = 0.5$ controls the balance between the hierarchical levels. Both losses use structural negative sampling to reinforce the separability between regions and between POIs. The output is the enriched *embeddings* $\mathbf{E}_{\text{HGI}} \in \mathbb{R}^{64}$, which capture both local and regional categorical relationships.

3.6. Embedding Integration

The three representation components are trained independently and concatenated to form the input of both MTLNet tasks:

$$\mathbf{E}_{\text{input}} = [\mathbf{E}_{\text{HGI}} \parallel \mathbf{E}_{\text{loc}} \parallel \mathbf{E}_{\text{time}}] \in \mathbb{R}^{192} \quad (4)$$

The choice of concatenation as the integration strategy preserves the individual information of each component without imposing premature interactions between the representations, delegating to MTLNet the task of learning how to combine these dimensions through its shared layers and FiLM modulation. The input dimensionality of ST-MTLNet ($\mathbb{R}^{192}$) is higher than that of the *baseline* ($\mathbb{R}^{64}$), a natural consequence of the decomposition into three components. However, the MTLNet architecture projects any input to the same shared latent space of dimension $d_{\text{shared}} = 256$ through the task-specific *encoders*, so that the capacity of the shared layers and task heads remains unchanged.

Two variants of the proposed model are evaluated, named ST-MTLNet (*Spatial-Temporal MTLNet*), each using a different spatial *encoder*. The *baseline* MTLNet [Silva et al. 2025] uses only the monolithic *embedding* $\mathbf{E}_{\text{DGI}} \in \mathbb{R}^{64}$ as input for both tasks. The proposed variants replace this representation with the concatenation of the decoupled *embeddings*, keeping the temporal (Time2Vec) and categorical (HGI) *encoders* fixed and varying the spatial *encoder* between SIREN and Sphere2Vec-M.

All models share the same hyperparameters of the MTLNet architecture, differing only in the input *embeddings*. This configuration allows isolating the effect of the representation strategy on performance, evaluating both the gain of the modular approach relative to the monolithic *baseline* and the impact of the choice of the spatial *encoder*.

4. Results

4.1. Experimental Setup

The performance of the models is evaluated using the Average F1-Score per category, reported as mean and standard deviation over 5 *folds* with a stratified split, using 80% of the data for training and 20% for validation, preserving the proportion between categories.

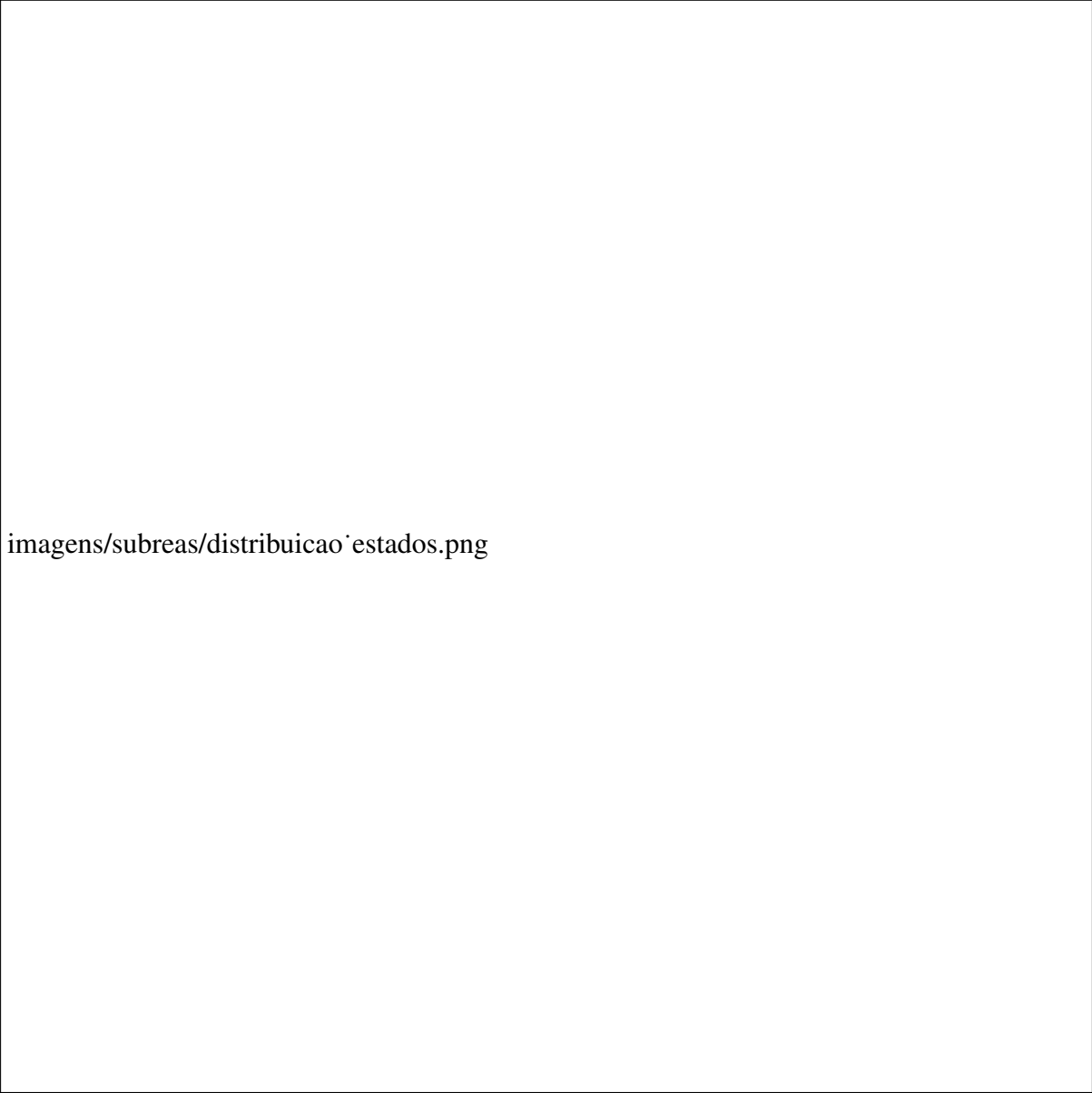
Table 1. Total number of *check-ins*, POIs, and users for Florida, California, and Texas.

States	<i>Check-ins</i>	Points of Interest	Users
Florida	990,518	65,009	20,301
California	2,535,573	148,314	36,106
Texas	3,355,419	135,570	37,522

The experiments were conducted with the Gowalla *dataset* [Liu et al. 2014] in the states of Florida, California, and Texas, whose statistics are presented in Table 1. The choice of these states is due to the high volume of *check-ins* and the presence of distinct urban patterns and spatial distributions, which allows evaluating the behavior of the models in scenarios with different levels of density and geographic dispersion.

Table 2. Average F1-Score (%) per model and state for the POI Category Classification task.

		MTLNet	ST-MTLNet _{SIREN}	ST-MTLNet _{Sphere2Vec-M}
Florida	Community	51.86 ± 0.73	70.00 ± 0.81	69.84 ± 0.98
	Entertainment	41.24 ± 1.83	64.45 ± 1.82	63.92 ± 1.61
	Food	55.47 ± 1.55	72.92 ± 0.59	73.22 ± 0.66
	Nightlife	32.59 ± 1.96	62.60 ± 1.96	62.44 ± 2.00
	Outdoors	47.71 ± 1.60	65.64 ± 1.73	66.35 ± 1.51
	Shopping	62.96 ± 0.62	77.48 ± 0.36	77.47 ± 0.72
	Travel	45.49 ± 1.20	64.89 ± 1.20	64.77 ± 1.54
California	Community	53.22 ± 0.37	70.21 ± 0.17	70.08 ± 0.44
	Entertainment	40.89 ± 1.32	63.60 ± 0.83	63.21 ± 0.76
	Food	60.68 ± 0.69	75.78 ± 0.39	75.74 ± 0.19
	Nightlife	26.81 ± 1.71	60.59 ± 1.35	60.78 ± 1.63
	Outdoors	51.59 ± 1.15	67.94 ± 1.52	67.37 ± 1.95
	Shopping	60.43 ± 1.01	76.84 ± 0.29	77.00 ± 0.33
	Travel	38.88 ± 1.32	63.59 ± 1.45	63.25 ± 0.99
Texas	Community	57.37 ± 0.69	73.45 ± 0.34	76.20 ± 0.46
	Entertainment	46.69 ± 0.46	64.09 ± 1.24	67.73 ± 1.43
	Food	54.31 ± 0.42	69.80 ± 0.80	73.10 ± 0.17
	Nightlife	34.44 ± 1.21	60.62 ± 1.32	64.57 ± 1.03
	Outdoors	45.99 ± 1.70	64.21 ± 0.42	68.72 ± 0.70
	Shopping	62.70 ± 0.29	76.40 ± 0.44	79.67 ± 0.14
	Travel	39.37 ± 1.29	57.53 ± 1.08	64.73 ± 0.72



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Figure 2. Spatial distribution of POIs of the Food (red) and Shopping (orange) categories in the densest sub-region of Florida, California, and Texas. The sub-regions were selected by density in a *grid*, with about 100 POIs per region.

4.2. POI Category Classification

The results of the POI Category Classification task are presented in Table 2. In the three evaluated states, the two variants of ST-MTLNet outperform the original MTLNet in all 21 category-state combinations, showing that replacing the monolithic *embedding* of DGI with decoupled representations consistently enriches the characterization of the POIs. The gains are particularly substantial in categories such as *Nightlife* and *Travel*. In *Nightlife*, for example, ST-MTLNet with Sphere2Vec-M reaches 62.44 ± 2.00 in Florida, 60.78 ± 1.63 in California, and 64.57 ± 1.03 in Texas, always well above the *baseline*. In *Travel*, the best results reach 64.89 ± 1.20 in Florida with SIREN, 63.59 ± 1.45 in California with SIREN, and 64.73 ± 0.72 in Texas with Sphere2Vec-M.

The improvements observed in *Food* and *Shopping* reinforce that the geographic location carries discriminative signals relevant to the semantic category of the POI. Figure 2 shows that these categories tend to form dense clusters in the analyzed sub-regions, favoring the capture of spatial patterns by the continuous *encoders*. Although the two proposed variants present close performance, a complementary behavior is observed: SIREN obtains more outstanding results in Florida and California, while Sphere2Vec-M stands out more frequently in Texas. Together, these results show that the modular approach consistently outperforms the monolithic paradigm of DGI in this task.

Table 3. Average F1-Score (%) per model and state for the Next-POI Prediction task.

		MTLNet	ST-MTLNet _{SIREN}	ST-MTLNet _{Sphere2Vec-M}
Florida	Community	34.33 ± 1.33	38.31 ± 0.90	37.98 ± 1.22
	Entertainment	25.34 ± 1.07	31.38 ± 1.72	31.24 ± 2.41
	Food	27.12 ± 2.79	41.91 ± 1.93	41.65 ± 3.15
	Nightlife	21.33 ± 1.61	23.32 ± 2.74	23.98 ± 1.94
	Outdoors	21.61 ± 0.99	21.29 ± 1.59	21.59 ± 1.91
	Shopping	39.62 ± 3.40	44.00 ± 2.04	44.66 ± 2.38
	Travel	64.47 ± 1.02	45.00 ± 1.10	44.93 ± 1.11
California	Community	32.14 ± 0.89	37.86 ± 0.20	37.80 ± 1.11
	Entertainment	19.38 ± 0.94	17.28 ± 1.73	17.01 ± 1.39
	Food	29.26 ± 1.67	50.42 ± 1.15	51.28 ± 0.57
	Nightlife	18.24 ± 0.53	16.34 ± 1.95	15.10 ± 2.70
	Outdoors	25.01 ± 0.81	24.84 ± 1.70	24.07 ± 2.46
	Shopping	38.41 ± 2.64	39.04 ± 1.72	37.82 ± 1.31
	Travel	46.05 ± 0.84	36.94 ± 1.70	37.82 ± 1.04
Texas	Community	34.22 ± 0.43	35.99 ± 0.93	36.97 ± 0.81
	Entertainment	25.56 ± 1.56	28.30 ± 1.37	27.59 ± 1.64
	Food	29.56 ± 4.10	47.48 ± 2.42	48.67 ± 0.66
	Nightlife	25.46 ± 1.13	26.06 ± 1.36	26.14 ± 2.66
	Outdoors	23.02 ± 1.03	24.62 ± 0.46	23.04 ± 1.44
	Shopping	38.79 ± 4.12	45.40 ± 1.30	43.18 ± 2.19
	Travel	29.71 ± 1.32	34.26 ± 0.90	33.05 ± 1.51

4.3. Next-POI Prediction

The results of the Next-POI Prediction task are presented in Table 3. Unlike classification, this task presents a more heterogeneous scenario, but still favorable to the proposed

variants: the spatio-temporal models outperform the original MTLNet in 16 of the 21 evaluated combinations. The largest gains occur in *Food*, a category in which both variants outperform the *baseline* in the three states. The best result is obtained by Sphere2Vec-M in Texas, with 48.67 ± 0.66 , while in California the same model reaches 51.28 ± 0.57 , outperforming DGI by a wide margin. Consistent improvements also appear in *Shopping* and *Community*, suggesting that the combination between continuous spatial information and temporal context favors the modeling of recurrent transitions between categories.

The co-location pattern shown in Figure 2 helps to interpret this behavior, since the presence of dense clusters of categories such as *Food* and *Shopping* favors frequent local transitions along the user’s trajectory. Even so, the *baseline* maintains an advantage in some cases, especially in *Travel* in Florida and California, in addition to specific categories in California, such as *Entertainment*, *Nightlife*, and *Outdoors*. These results suggest that long-distance movements and topological relationships between regions are still better represented by DGI, whose spatial graph captures structural connections that the coordinate-based *encoders* do not model directly. Even with this limitation, the general performance indicates that the proposed representations are more effective in most sequential prediction scenarios.

5. Conclusion and Future Work

This work investigated whether replacing the monolithic *embedding* of DGI with decoupled and specialized representations could produce a base more suitable for the tasks of POI Category Classification and Next-POI Prediction. The results obtained in the three evaluated states indicate that it can. By incorporating a continuous spatial *encoder*, a temporal *encoder* (Time2Vec), and a hierarchical categorical *encoder* (HGI), ST-MTLNet consistently outperforms the *baseline* based only on DGI.

In the POI Category Classification task, the proposed variants outperform MTLNet in all 21 category-state combinations, with average gains of 20 to 24 percentage points. In the Next-POI Prediction task, the performance is also mostly favorable to ST-MTLNet, which wins in 16 of the 21 evaluated combinations, with emphasis on categories such as *Food*. Taken together, these results show that decomposing the representation into spatial, temporal, and categorical components allows capturing patterns that DGI, by itself, does not model sufficiently.

The comparison between SIREN and Sphere2Vec-M also shows that there is no single universally superior spatial *encoder*. SIREN presents better performance more frequently in Florida and California, while Sphere2Vec-M stands out in Texas, suggesting that the suitability of the *encoder* depends on the geographic distribution of the POIs in each territory. Thus, the main gain of this work lies not only in the choice of a specific spatial architecture, but in the very adoption of decoupled representations as an alternative to the monolithic paradigm of DGI.

A relevant limitation remains in the *Travel* category of the Next-POI Prediction task, in which the original MTLNet still obtains the best results in part of the scenarios. This suggests that long-distance movements and topological relationships between regions are still better captured by the graph structure used by DGI. In addition, since the three proposed components are used together, this work does not isolate the individual contribution of each *encoder*, which restricts a more detailed analysis of the relative weight of each source of information.

As future work, we highlight the investigation of more flexible multi-task architectures, the exploration of more sophisticated fusion strategies between the *encoders*, and the combination of graph-based representations and continuous coordinate encodings. These directions

may broaden the observed gains and help reduce the limitations still present in categories such as *Travel*.

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